Rama Krishna Kamma

kammaramakrishna12@gmail.com | +1 (903) 246-2287 | LinkedIn | Portfolio

SUMMARY:

Data Engineer with 6+ years of experience building production-grade ETL/ELT pipelines and cloud data architectures across Healthcare, Supply Chain, and Freight domains. Delivered $5M+ in business savings by engineering freight optimization pipelines, carrier scoring models, and real-time operational datasets on Databricks. Expert in PySpark, Python, SQL, and Azure with deep hands-on experience in Delta Lake, Medallion Architecture, and Power BI data layers. Certified Databricks Data Engineer Professional with a strong track record of improving data quality, reducing manual workflows, and delivering trusted data at scale.

SKILLS SUMMARY:

- **PySpark:** DataFrames, Structured Streaming, Broadcast Joins, Spark SQL, Catalyst Optimizer, Tungsten Engine, Partition Tuning, Schema Enforcement, Delta Lake Writes, Spark UI & DAG Analysis, Data Skew Handling.
- **Databricks:** Lakehouse, Unity Catalog, Delta Live Tables (DLT), Genie, Asset Bundles, Agentic AI, Serverless SQL
- **Data Engineering:** Azure Data Factory (ADF), Apache Airflow, Informatica, Kafka, Delta Lake, Snowflake, Microsoft Fabric.
- **Databases & Storage:** MySQL, SQL Server, Oracle, MongoDB, Azure SQL, ADLS Gen2, Azure Synapse, AWS S3, GCS, Dataverse, Azure Synapse, Azure Resource Graph.
- **Programming Languages & Querying:** Python, SQL (Advanced Tuning, Window Functions, Stored Procedures), KQL, DAX, Java, C/C++
- **AI & Visualization:** Microsoft Power BI, AI/BI Databricks Dashboards, Tableau, GenAI (RAG, LLMs, Embeddings, Vector Search), Semantic Modeling, Genie Code.
- **DevOps & Other tools:** Azure DevOps, Jenkins, Git, GitHub, GitLab, Bicep, ARM Templates, Power Platform (PowerApps, Automate), JIRA, Confluence, Agile (Scrum/Kanban), Azure Data Explorer, Postman.

EDUCATION & CERTIFICATIONS:

- Master of Science (Computer and Information Sciences) from Texas A&M University - Commerce
- Databricks Certified Data Engineer Professional
- Databricks Certified Data Engineer Associate
- Azure Administrative Associate (AZ-104)
- Azure Power BI Data Analyst Associate (PL-300)
- Azure Power Platform Fundamentals (PL-900)
- Azure Fundamentals (AZ-900)

PROFESSIONAL EXPERIENCE:

Optum — **Mar 2026 – Present**

Data Engineer — **Minneapolis, Minnesota**

- Migrated legacy **Informatica MCP** workflows to Databricks **PySpark**, replacing single-node ETL mappings with distributed transformations improving scalability, maintainability, and processing performance.
- Built **coverage interval splitting** pipeline in PySpark to clamp member date ranges into monthly reporting windows and distribute premium values proportionally using window functions and date arithmetic.
- Engineered **HHS enrollment** data pipeline processing subscriber and dependent records through one-day premium checks, pharmacy/medical claims verification, soft exclusions (DOB/gender anomalies), and hard exclusions (invalid premiums).
- Developed **EPAI code** assignment and **age recalculation** logic (codes 001 / 021028 / special) with CPIN validation across reporting year migrations for accurate HHS Risk Adjustment submissions.
- Partnered with compliance and business stakeholders to translate **CMS/HHS regulatory** requirements into validated PySpark transformations, ensuring timely and accurate annual EDGE Server submissions.

Radius Recycling — **Nov 2024 – Mar 2026**

Data Engineer — **Portland, Oregon**

- Designed and built **PySpark transformation** pipelines using DataFrames with complex multi-source joins (filter, groupBy, aggregations, window functions) to process freight, carrier, and supply chain datasets — similar to TMS and carrier platform integrations — driving **$5M+ in annual savings** and **12% cost reductions**.
- Implemented **Structured Streaming** with PySpark to process real-time carrier event data applying **watermarking**, aggregations, Delta Lake with schema enforcement for reliable, low-latency pipeline delivery.
- Authored reusable PySpark functions for custom business logic including freight cost normalization, carrier

scoring algorithms, lane-level pricing margin analysis, and anomaly flagging reducing repetitive transformation code across multiple pipeline jobs.

- Optimized Spark workloads using **broadcast joins**, **Catalyst optimizer hints**, and partition tuning eliminating shuffle-heavy operations and reducing job execution time significantly across datasets.
- Recovered **$40K in overpayments** by building lane-wise cost variance and pricing margin models with Delta Lake writes, delivering curated Power BI dashboards for operations and pricing teams with automated weekly stakeholder reporting via **Power Automate**.
- Reduced annual reporting costs by **75% ($24K to $6K)** by re-architecting ELT pipelines using PySpark on Databricks implementing Bronze/Silver/Gold **Medallion Architecture** with enforced data quality, schema evolution, and lineage tracking.
- Built production-grade **Spark SQL** transformation layers using CTEs and Catalyst optimizer-aware query patterns ingesting from MySQL, Oracle, and REST APIs via **Databricks Asset Bundles** and Git integration.
- Developed Databricks **Agentic AI assistant** for natural language Q&A on compliance documents and ML models for freight price forecasting and carrier rate prediction; built **GenAI RAG** solutions using Unity Catalog governance and Databricks Genie.
- Led Root-Cause Analysis (RCA) and Post-Incident Reviews (PIR) for pipeline failures; utilized KQL and **Azure Log Analytics** to monitor Spark job telemetry, develop automated alerting, and proactively resolve anomalies.
- Mentored junior engineers on PySpark best practices, Delta Lake schema management, and pipeline optimization; defined data quality standards and automation frameworks to reduce manual data intervention; optimized Data Warehouse costs by migrating dev workloads to Databricks **Serverless SQL**.

Texas A&M University **Sep2022 – May 2024**

Business Intelligence Developer – Data Analytics and Data Visualization **Commerce, TX**

- Reduced manual reporting effort by **80%** by building PySpark based automation workflows using Azure Databricks and ADF, enabling faster access to critical operational insights.
- Reduced ad-hoc data requests by 70% by developing **self-serve BI** tools and shared datasets for class engagement and peak-hour analysis, enabling leadership with actionable insights.
- Boosted **member engagement by 20%** through activity trend dashboards that surfaced underutilized time slots and drove new event recommendations.
- Enforced Zero Trust access principles, **RBAC**, and Azure Sentinel monitoring in collaboration with IT security, ensuring strict data privacy governance.
- Optimized **Power BI Vertipaq storage** and implemented parameterized **data refreshes**, significantly reducing dashboard load times.

Tata Consultancy Services **Dec 2021 – Aug 2022**

Data Engineer – PySpark and Cloud Integration **IN**

- Engineered **Medallion Architecture** (Bronze/Silver/Gold) using PySpark DataFrames and Delta Lake on Databricks writing modular, reusable transformation logic with schema enforcement and automated data **quality checks**, boosting query performance by **40%**.
- Reduced query latency by 25% by implementing **Z-Ordering**, data skipping, broadcast joins, and **Spark SQL** Catalyst optimizer hints eliminating expensive shuffle operations across distributed PySpark workloads.
- Built PySpark Structured Streaming pipelines with aggregations and Delta Lake ACID writes enabling reliable **incremental data processing** with schema evolution and automatic checkpoint recovery.
- Reduced incident response time by **35%** by utilizing **Spark UI** metrics, DAG visualization, and automated log analysis to identify executor bottlenecks, **skewed partitions**, and GC pressure resolving anomalies proactively.
- Established **CI/CD** pipelines via **Azure DevOps**, achieving a 15% reduction in cloud compute spend through optimized Spark cluster sizing, job scheduling, and partition tuning.
- Secured sensitive **PII data** using **Azure Key Vault** and enforced **Unity Catalog** data masking, lineage tracking, and column-level security policies across all PySpark pipelines.
- Developed dimensional models using **SCD Type 1/2** with PySpark merge operations and automated anomaly detection to ensure high data integrity across warehousing layers; structured and curated clean datasets for downstream machine learning and AI initiatives.

Client: Schnitzer Steel **Nov 2020 – Nov 2021**

Data Engineer – Spark & ETL Optimization **IN**

- Managed lift-and-shift migration of on-premises Hadoop workloads to Databricks rewriting legacy MapReduce jobs as optimized PySpark DataFrames with partition pruning, **predicate pushdown**, and broadcast join

strategies to maximize performance.

- Maintained Spark Streaming and **batch pipelines** processing safety and transportation data implementing High Availability for ADLS Gen2 through replication and failover to minimize latency for real-time analytics.
- Achieved **40%** improvement in data processing through deep PySpark performance tuning analyzing **Spark UI DAGs**, resolving data skew with **salting** techniques, right-sizing executors, and optimizing shuffle partitions.
- Streamlined release management across ADF environments using **ARM** and **Bicep** templates for automated code promotion and infrastructure deployment to enhance live site reliability.
- Designed Power BI reporting solutions for route optimization, carrier performance, and equipment downtime analytics — delivering curated data models and reusable views translating PySpark-processed datasets into stakeholder-ready dashboards.

UrbanJet Hospitality Services, Hyderabad, India **Jan 2019 – Jun 2020**
Data Analyst Intern

- Designed and optimized semantic models using **Power Query** and **DAX** with Row-Level Security (RLS); built **SSIS** packages for streamlined data integration across cloud and on-premises systems, ensuring high data integrity.
- Conducted **exploratory data analysis (EDA)** to uncover trends and anomalies, translating findings into intuitive Power BI reports and technical documentation accessible to non-technical stakeholders.
- Demonstrated strong problem-solving skills and communication skills by resolving complex data challenges and refining reporting processes, effectively **prioritizing** multiple tasks to meet strict project deadlines.
- Collaborated with **cross-functional teams** and led retrospective sessions to capture best practices, utilizing strong written and verbal communication to enhance team productivity and workflow efficiency.